

Back to multinomial (Agresti 1.5)

Recap: $X_{k \times 1} \sim \text{Multinomial}_k(n, \underline{p})$

$$\underline{p}_{k \times 1} \quad p_i \geq 0, \quad \sum_{i=1}^k p_i = 1$$

$\underline{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_k \end{pmatrix}$ $X_i = \# \text{ of times } i\text{th outcome happens out of } n \text{ trials.}$

$$\sum_{i=1}^k X_i = n.$$

• pmf $P(\underline{X} = \underline{x}) = \begin{cases} \frac{n!}{x_1! \dots x_k!} p_1^{x_1} \dots p_k^{x_k}, & \sum_{i=1}^k x_i = n \\ 0 & \text{o.w.} \end{cases}$

• $\hat{p}_{MLE} = \frac{1}{n} \underline{x}$

• $E \hat{p}_{MLE} = p \quad \text{Var } \hat{p} = \frac{1}{n} \Sigma$

$$\Sigma_{ii} = p_i(1 - p_i)$$

$$\Sigma_{ij} = -p_i p_j \quad \text{if } i \neq j$$

• $\hat{p} \xrightarrow[p]{a.s.} p \quad \text{SLLN}$

• $\frac{\sqrt{n}(\hat{p}_i - p_i)}{\sqrt{p_i(1 - p_i)}} \Rightarrow N(0, 1). \quad \forall i = 1, \dots, k.$

Test hypothesis $H_0: p = p_0$

$H_a: p \neq p_0$

~~$\sqrt{n}(\hat{p} - p)$~~

$$X = \sum_{j=1}^n Y_j$$

$$Y_j = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \rightarrow \text{correct outcome}$$

Multivariate CLT. $\begin{cases} y_j \text{ are indep identically distributed} \\ E y_j = p \\ \text{Var } y_j = \Sigma \end{cases}$ $E y_1 = 1P(\text{1st trial gives outcome 1}) + 0P(\text{... don't see ...}) = p_1$

$$\sqrt{n} \Sigma^{-1/2} (X/n - p) \Rightarrow N(0, I)$$

$$\frac{\sqrt{n}(\bar{x} - \mu)}{\sqrt{\sigma^2}} \Rightarrow N(0, 1)$$

$$\Sigma = \begin{bmatrix} p_1(1-p_1) & -p_1 p_2 & \dots & -p_1 p_k \\ -p_1 p_2 & p_2(1-p_2) & \dots & -p_2 p_k \\ \vdots & \vdots & \ddots & \vdots \\ -p_k p_1 & -p_k p_2 & \dots & p_k(1-p_k) \end{bmatrix}$$

Σ is singular.

$$1^T \Sigma = 0^T$$

$$\begin{aligned} & -p_1 p_2 - p_3 p_2 - \dots - p_k p_2 + p_2(1-p_2) \\ & = -p_2(p_1 + p_3 + \dots + p_k) + p_2(1-p_2) = 0 \end{aligned}$$

Rephrase the problem

$$p^* = (p_1, \dots, p_{k-1})^T$$

$$H_0: p^* = p_0^*$$

$$H_a: p^* \neq p_0^*$$

$$X^* = \begin{pmatrix} X_1 \\ \vdots \\ X_{k-1} \end{pmatrix}$$

$$X_k = n - \sum_{i=1}^{k-1} X_i$$

$$E X^* = p^*$$

$$\text{Var } X^* = \Sigma^*$$

$$\Sigma^* = \begin{bmatrix} \Sigma^* & 0 \\ 0 & 0 \end{bmatrix}$$

$$\sqrt{n} \Sigma^{*-1/2} (X^*/n - p^*) \Rightarrow N_{k-1}(0, I)$$

$$Z_n^* \Rightarrow Z^*$$

$$Z^{*T} Z^* \sim \chi^2_{k-1}$$

$$Z^{*T} Z^* \Rightarrow \chi^2_{k-1}$$

$$Z_n^T Z_n \Rightarrow \chi_{k-1}^2$$

Claim: $n(\bar{X} - p^*)^T \Sigma^{-1} (\bar{X} - p^*) \Rightarrow \chi_{k-1}^2$

Claim: $\Sigma^{-1} = \text{diag}(\frac{1}{p^*}) + \frac{1}{p_k} \mathbf{1}\mathbf{1}^T \quad (A)$

Prf: $A \Sigma^* = \Sigma^* A = I$ (to show)

$$\begin{bmatrix} \frac{1}{p_1} & \dots & \frac{1}{p_k} & \dots & \frac{1}{p_k} \\ \vdots & & \vdots & & \vdots \\ \frac{1}{p_k} & \dots & \frac{1}{p_k} & \dots & \frac{1}{p_k} \end{bmatrix}$$

$$\Sigma^* = \text{diag}(p^*) - p^* p^{*T}$$

$$\begin{aligned} A \Sigma^* &= \left[\text{diag}(\frac{1}{p^*}) + \frac{1}{p_k} \mathbf{1}\mathbf{1}^T \right] [\text{diag}(p^*) - p^* p^{*T}] \\ &= \text{diag}(\frac{1}{p^*}) \text{diag}(p^*) + \frac{1}{p_k} \mathbf{1}\mathbf{1}^T \text{diag}(p^*) - \frac{\text{diag}(\frac{1}{p^*}) p^* p^{*T}}{\frac{1}{p_k} \mathbf{1}\mathbf{1}^T p^* p^{*T}} \\ &= I + \frac{1}{p_k} \mathbf{1} p^{*T} - \mathbf{1} p^{*T} - \frac{1}{p_k} \mathbf{1} (1 - p_k) p^{*T} \end{aligned}$$

$$\begin{aligned} (1 \dots 1) \begin{bmatrix} p_1 & \dots & p_k & \dots & p_k \\ \vdots & & \vdots & & \vdots \\ p_k & \dots & p_k & \dots & p_k \end{bmatrix} &= (1 \dots 1 \dots 1) \\ &= I + \mathbf{1} p^{*T} \left(\frac{1}{p_k} - 1 - \frac{1 - p_k}{p_k} \right) = I \end{aligned}$$

Show $\Sigma^* A = I$.

Test of hypothesis $H_0: p^* = p_0$,

under H_0 , $B: n(\bar{X} - p_0)^T A (\bar{X} - p_0) \xrightarrow{\text{as } n \rightarrow \infty} \chi_{k-1}^2$

$$A = \text{diag}(\frac{1}{p_0}) - \frac{1}{p_k} \mathbf{1}\mathbf{1}^T$$

$$A = \text{diag}(\frac{1}{p_0^*}) - \frac{1}{p_k^*} \mathbf{1}\mathbf{1}^T$$

B_n is a statistic

Reject H_0 according to quantiles of χ_{k-1}^2

Right tailed cut-off

Reject for large values of B_n .

Claim 2: $B_n = \sum_{i=1}^k \frac{(x_i - np_i)^2}{np_i} = \sum_{i=1}^k \frac{(\text{obs}_i - \text{exp}_i)^2}{\text{exp}_i}$

k cells. $x_i \rightarrow$ observed count

$np_i \rightarrow$ expected count

MCLT.